

Rainbow connections of graphs – A survey*

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Abstract

The concept of rainbow connection was introduced by Chartrand et al. in 2008. It is fairly interesting and recently quite a lot papers have been published about it. In this survey we attempt to bring together most of the results and papers that dealt with it. We begin with an introduction, and then try to organize the work into five categories, including (strong) rainbow connection number, rainbow k -connectivity, k -rainbow index, rainbow vertex-connection number, algorithms and computational complexity. This survey also contains some conjectures, open problems or questions.

Keywords: rainbow path, (strong) rainbow connection number, rainbow k -connectivity, k -rainbow index, rainbow vertex-connection number, algorithm, computational complexity

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1 Introduction

1.1 Motivation and definitions

Connectivity is perhaps the most fundamental graph-theoretic subject, both in combinatorial sense and the algorithmic sense. There are many elegant and powerful results on connectivity in graph theory. There are also many ways to strengthen the connectivity concept, such as requiring hamiltonicity, k -connectivity, imposing bounds on the diameter, and so on. An interesting way to strengthen the connectivity requirement, the rainbow connection, was introduced by Chartrand, Johns, McKeon and Zhang [12] in 2008, which is restated as follows:

This new concept comes from the communication of information between agencies of government. The Department of Homeland Security of USA was created in 2003 in response

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to the weaknesses discovered in the transfer of classified information after the September 11, 2001 terrorist attacks. Ericksen [25] made the following observation: An unanticipated aftermath of those deadly attacks was the realization that law enforcement and intelligence agencies couldn't communicate with each other through their regular channels, from radio systems to databases. The technologies utilized were separate entities and prohibited shared access, meaning that there was no way for officers and agents to cross check information between various organizations.

While the information needs to be protected since it relates to national security, there must also be procedures that permit access between appropriate parties. This two-fold issue can be addressed by assigning information transfer paths between agencies which may have other agencies as intermediaries while requiring a large enough number of passwords and firewalls that is prohibitive to intruders, yet small enough to manage (that is, enough so that one or more paths between every pair of agencies have no password repeated). An immediate question arises: What is the minimum number of passwords or firewalls needed that allows one or more secure paths between every two agencies so that the passwords along each path are distinct?

This situation can be modeled by graph-theoretic model. Let G be a nontrivial connected graph on which an edge-coloring $c : E(G) \rightarrow \{1, 2, \dots, n\}$, $n \in \mathbb{N}$, is defined, where adjacent edges may be colored the same. A path is *rainbow* if no two edges of it are colored the same. An edge-coloring graph G is *rainbow connected* if any two vertices are connected by a rainbow path. An edge-coloring under which G is rainbow connected is called a *rainbow coloring*. Clearly, if a graph is rainbow connected, it must be connected. Conversely, any connected graph has a trivial edge-coloring that makes it rainbow connected; just color each edge with a distinct color. Thus, we define the *rainbow connection number* of a connected graph G , denoted by $rc(G)$, as the smallest number of colors that are needed in order to make G rainbow connected [12]. A rainbow coloring using $rc(G)$ colors is called a *minimum rainbow coloring*. So the question mentioned above can be modeled by means of computing the value of rainbow connection number. By definition, if H is a connected spanning subgraph of G , then $rc(G) \leq rc(H)$. For a basic introduction to the topic, we refer the readers to Chapter 11 in [16].

In addition to regarding as a natural combinatorial measure and its application for the secure transfer of classified information between agencies, rainbow connection number can also be motivated by its interesting interpretation in the area of networking [10]: Suppose that G represents a network (e.g., a cellular network). We wish to route messages between any two vertices in a pipeline, and require that each link on the route between the vertices (namely, each edge on the path) is assigned a distinct channel (e.g. a distinct frequency). Clearly, we want to minimize the number of distinct channels that we use in our network. This number is precisely $rc(G)$.

Let c be a rainbow coloring of a connected graph G . For any two vertices u and v of G , a *rainbow $u - v$ geodesic* in G is a rainbow $u - v$ path of length $d(u, v)$, where $d(u, v)$ is the distance between u and v in G . A graph G is *strong rainbow connected* if there exists a

rainbow $u-v$ geodesic for any two vertices u and v in G . In this case, the coloring c is called a *strong rainbow coloring* of G . Similarly, we define the *strong rainbow connection number* of a connected graph G , denoted $src(G)$, as the smallest number of colors that are needed in order to make G strong rainbow connected [12]. Note that this number is also called the *rainbow diameter number* in [10]. A strong rainbow coloring of G using $src(G)$ colors is called a *minimum strong rainbow coloring* of G . Clearly, we have $diam(G) \leq rc(G) \leq src(G) \leq m$, where $diam(G)$ denotes the diameter of G and m is the size of G .

In a rainbow coloring, we only need to find one rainbow path connecting any two vertices. So there is a natural generalization: the number of rainbow paths between any two vertices is at least an integer k with $k \geq 1$ in some edge-coloring. A well-known theorem of Whitney [55] shows that in every κ -connected graph G with $\kappa \geq 1$, there are k internally disjoint $u-v$ paths connecting any two distinct vertices u and v for every integer k with $1 \leq k \leq \kappa$. Similar to rainbow coloring, we call an edge-coloring a *rainbow k -coloring* if there are at least k internally disjoint $u-v$ paths connecting any two distinct vertices u and v . Chartrand, Johns, McKeon and Zhang [13] defined the *rainbow k -connectivity* $rc_k(G)$ of G to be the minimum integer j such that there exists a j -edge-coloring which is a *rainbow k -coloring*. A rainbow k -coloring using $rc_k(G)$ colors is called a *minimum rainbow k -coloring*. By definition, $rc_k(G)$ is the generalization of $rc(G)$ and $rc_1(G) = rc(G)$ is the rainbow connection number of G . By coloring the edges of G with distinct colors, we see that every two vertices of G are connected by k internally disjoint rainbow paths and that $rc_k(G)$ is defined for every $1 \leq k \leq \kappa$. So $rc_k(G)$ is well-defined. Furthermore, $rc_k(G) \leq rc_j(G)$ for $1 \leq k \leq j \leq \kappa$. Note that this new defined *rainbow k -connectivity* computes the number of *colors*, this is distinct with connectivity (edge-connectivity) which computes the number of internally (edge) disjoint *paths*. We can also call it *rainbow k -connection number*.

Now we introduce another generalization of rainbow connection number by Chartrand, Okamoto and Zhang [15]. Let G be an edge-colored nontrivial connected graph of order n . A tree T in G is a *rainbow tree* if no two edges of T are colored the same. Let k be a fixed integer with $2 \leq k \leq n$. An edge coloring of G is called a *k -rainbow coloring* if for every set S of k vertices of G , there exists a rainbow tree in G containing the vertices of S . The *k -rainbow index* $rx_k(G)$ of G is the minimum number of colors needed in a k -rainbow coloring of G . A k -rainbow coloring using $rx_k(G)$ colors is called a *minimum k -rainbow coloring*. Thus $rx_2(G)$ is the rainbow connection number $rc(G)$ of G . It follows, for every nontrivial connected graph G of order n , that $rx_2(G) \leq rx_3(G) \leq \dots \leq rx_n(G)$.

The above four new graph-parameters are all defined in edge-colored graphs. Krivelevich and Yuster [36] naturally introduced a new parameter corresponding to rainbow connection number which is defined on vertex-colored graphs. A vertex-colored graph G is *rainbow vertex-connected* if any two vertices are connected by a path whose *internal* vertices have distinct colors. A vertex-coloring under which G is rainbow vertex-connected is called a *rainbow vertex-coloring*. The *rainbow vertex-connection number* of a connected graph G , denoted by $rvc(G)$, is the smallest number of colors that are needed in order to make G rainbow vertex-connected. The minimum rainbow vertex-coloring is defined sim-